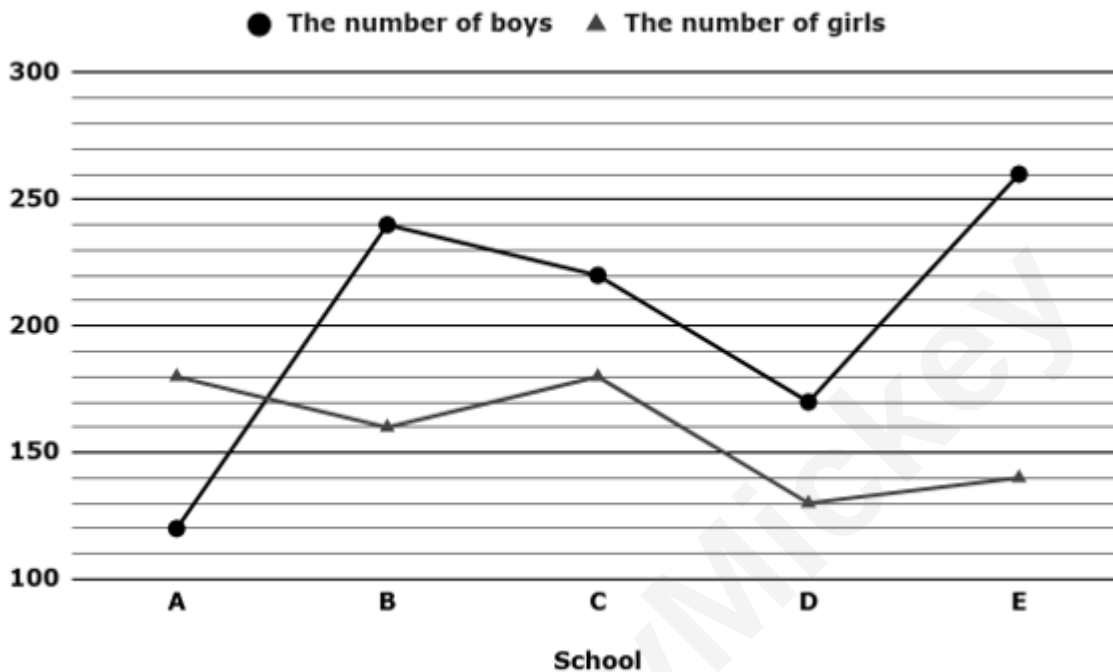


1. Questions

Study the following information carefully and answer the questions given below.

The line graph given below shows the number of boys and girls in five different schools namely A, B, C, D and E.



Note: The total number of students = The number of boys + The number of girls.

Out of the total students in School B, 45% are participating in an inter-school sports event. If 60% of these participating students are boys, then find the number of girls in School B who are NOT participating in the sports event.

- 76
- 88
- 92
- 84
- 72

2. Questions

If the total number of students in School F is 25% more than the total number of students in School C, and the ratio of boys to girls in School F is 3 : 2, find the difference between the number of boys in School F and the number of girls in School D.

- 170
- 160
- 180
- 150

e. 140

3. Questions

The number of boys in School B is what percentage of the number of boys in School A?

- a. 100%
- b. 150%
- c. 200%
- d. 180%
- e. 240%

4. Questions

What is the average number of girls in Schools A, B, and E?

- a. 180
- b. 140
- c. 150
- d. 120
- e. 160

5. Questions

What is the ratio of the number of boys in School B to the number of girls in School E?

- a. 13 : 11
- b. 11 : 6
- c. 5 : 9
- d. 12 : 7
- e. 7 : 3

6. Questions

Study the following information carefully and answer the questions given below.

The table chart given below shows the number of female employees working in five different companies namely P, Q, R, S and T along with the difference between the number of male employees and female employees working in each company.

Company	The number of female employees	The difference between female and male employees
P	128	32
Q	132	28
R	144	46
S	156	24
T	115	35

Notes:

- (i) The total number of employees = The number of male employees + The number of female employees.
- (ii) The number of male employees in each department is greater than the number of female employees.

If company U has 25% more male employees than company P and the number of female employees in company U is equal to the number of female employees in company S, then what is the total number of employees working in company U?

- a. 432
- b. 356
- c. 448
- d. 332
- e. 456

7. Questions

Find the ratio of the difference between the number of female employees in companies R and Q to the difference between the number of male employees in companies S and Q.

- a. 3 : 5
- b. 2 : 1
- c. 5 : 7
- d. 4 : 5
- e. 7 : 9

8. Questions

The number of male employees in company R is what percentage more or less than the number of

male employees in company T?

- a. 37.5% more
- b. 33.33% less
- c. 24.12% more
- d. 13.33% less
- e. 26.67% more

9. Questions

What is the difference between the number of female employees in companies P and T?

- a. 7
- b. 11
- c. 9
- d. 13
- e. 6

10. Questions

If the average number of employees in companies P and T is $2s$, then find the value of s .

- a. 143.25
- b. 276.5
- c. 138.25
- d. 243.5
- e. 148.25

11. Questions

Study the following information carefully and answer the questions given below.

The distance between City P and City Q is 350 km. Three different cars A, B and C travel between these two cities. Car A takes 7 hours to cover the entire distance from City P to City Q. The speed of Car B is 20% more than the speed of Car A. The speed of Car C is 25% more than the speed of Car B.

If Car A increases its speed by 20 km/h, how much less time will it take to cover the entire distance between City P and City Q compared to its original time?

- a. 2 hours
- b. 3 hours
- c. 1.5 hours
- d. 4 hours

e. 2.5 hours

12. Questions

What is the ratio of the time taken by Car A to the time taken by Car B to cover a distance of 450 km?

- a. 3 : 2
- b. 5 : 4
- c. 6 : 5
- d. 5 : 2
- e. 8 : 7

13. Questions

What is the total time taken by Car C to cover the entire distance between City P and City Q?

- a. 15/7 hours
- b. 14/5 hours
- c. 15/8 hours
- d. 13/7 hours
- e. 14/3 hours

14. Questions

A sum invested at simple interest amounts to Rs. 4800 in 3 years and Rs. 5600 in 5 years. Find the rate of interest per annum.

- a. 11.11%
- b. 9.09%
- c. 12.5%
- d. 13.33%
- e. 5.45%

15. Questions

90 liters of a mixture contains milk and water in the ratio of 7 : 2. If 27 liters of the mixture is taken out and the vessel is then filled with 21 liters of pure water, what is the ratio of milk and water in the final solution?

- a. 4 : 3
- b. 7 : 5
- c. 2 : 1

d. 6 : 5

e. 9 : 8

16. Questions

An article was sold by a shopkeeper for Rs. 680 after offering a discount of 15% on its marked price. If the ratio of the marked price to the cost price of the article is 4 : 3, what is the cost price of the article?

a. Rs. 500

b. Rs. 800

c. Rs. 600

d. Rs. 700

e. Rs. 400

17. Questions

Two pipes A and B can fill a tank in 15 minutes and 18 minutes respectively. If both pipes are opened together, how much time will it take to fill the tank?

a. 8 $\frac{2}{9}$ minutesb. 7 $\frac{2}{11}$ minutesc. 8 $\frac{4}{11}$ minutesd. 8 $\frac{2}{11}$ minutese. 3 $\frac{2}{13}$ minutes**18. Questions**

P, Q, and R entered into a partnership investing Rs. 18000, Rs. 21000 and Rs. 24000 respectively. If the total profit at the end of the year is Rs. 63000, find the share of Q.

a. Rs. 23000

b. Rs. 22000

c. Rs. 26000

d. Rs. 24000

e. Rs. 21000

19. Questions

The ratio of the present ages of A and B is 4 : 5, respectively. After 6 years, the ratio of their ages will be 5 : 6. Find the present age of C, who is 4 years older than the average age of A and B.

a. 31 years

- b. 28 years
- c. 35 years
- d. 24 years
- e. 39 years

20. Questions

The breadth of a rectangle is 50% of its length. If the length of the rectangle is equal to the side of a square whose area is 484 cm^2 , find the perimeter of the rectangle.

- a. 44 cm
- b. 66 cm
- c. 33 cm
- d. 88 cm
- e. 77 cm

21. Questions

What value should come in the place of (?) in the following questions.

$13\% \text{ of } 400 + ? = 23^2 - 17 * 12 + 19$

- a. 392
- b. 232
- c. 312
- d. 292
- e. 342

22. Questions

$5/14 * 980 + 150 = ? + 15 * 24$

- a. 140
- b. 110
- c. 120
- d. 130
- e. 150

23. Questions

$19^2 + \sqrt{1296} - 150 = ? + 16.66\% \text{ of } 72$

- a. 231
- b. 243
- c. 235
- d. 253
- e. 256

24. Questions

$35\% \text{ of } 800 + 16^2 + ? = 5/8 * 1024$

- a. 117
- b. 91
- c. 65
- d. 113
- e. 104

25. Questions

$7/9 * 729 + 45 * 12 = ? + 145$

- a. 962
- b. 845
- c. 943
- d. 895
- e. 917

26. Questions

What approximate value will come in the place of (?) in the following questions?

$35.11\% \text{ of } 979.99 + 935.99 / 12.99 = ?$

- a. 295
- b. 415
- c. 335
- d. 495
- e. 285

27. Questions

$4.99 * 79.99 + 3.99 * 59.99 = ? + 64.99\% \text{ of } 399.99$

- a. 380
- b. 420
- c. 310
- d. 480
- e. 350

28. Questions

$44.99\% \text{ of } 199.99 = ? - 19.99 * 9.99 + 24.99\% \text{ of } 799.99$

- a. 70
- b. 80
- c. 90
- d. 60
- e. 50

29. Questions

$39.99 * 14.99 - 24.99 * 11.99 = ? - 13.99 * 15.01$

- a. 610
- b. 210
- c. 310
- d. 510
- e. 410

30. Questions

$12.21\% \text{ of } 799.99 + 87.29\% \text{ of } 599.98 = ?$

- a. 673
- b. 518
- c. 692
- d. 529
- e. 618

31. Questions

What value will come in the place of (?) in the following number series?

12, 12, 18, 36, 90, ?

- a. 270
- b. 240
- c. 180
- d. 150
- e. 210

32. Questions**5, 6, 14, 45, ?, 925**

- a. 112
- b. 184
- c. 162
- d. 198
- e. 148

33. Questions**5, 11, 23, 47, 95, ?**

- a. 171
- b. 181
- c. 191
- d. 161
- e. 151

34. Questions**32, 36, 45, 61, 86, ?**

- a. 111
- b. 144
- c. 133
- d. 122
- e. 155

35. Questions**18, 27, 45, 72, 108, ?**

- a. 148

- b. 151
- c. 139
- d. 142
- e. 153

36. Questions

Find the wrong number in the following number series.

4, 12, 36, 108, 324, 970

- a. 12
- b. 970
- c. 36
- d. 108
- e. 4

37. Questions

20, 30, 60, 155, 450, 1575

- a. 155
- b. 20
- c. 1575
- d. 60
- e. 450

38. Questions

50, 52, 49, 54, 47, 59

- a. 49
- b. 50
- c. 52
- d. 47
- e. 59

39. Questions

43, 64, 87, 112, 140, 168

- a. 112

- b. 43
- c. 140
- d. 64
- e. 168

40. Questions**5, 10, 30, 120, 600, 3500**

- a. 120
- b. 5
- c. 10
- d. 3500
- e. 600

41. Questions

The following questions contain two equations as I and II. You have to solve both equations and determine the relationship between them and give answer as,

I). $x^2 - 27x + 182 = 0$

II). $y^2 - 21y + 108 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

42. Questions

I). $x^2 - 16x + 63 = 0$

II). $y^2 - 30y + 209 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $x^2 - 28x + 195 = 0$

II). $y^2 - 23y + 130 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

44. Questions

I). $x^2 - 31x + 228 = 0$

II). $y^2 - 40y + 399 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

45. Questions

I). $x^3 = 216$

II). $y^2 - 12y + 36 = 0$

- a. $x > y$
- b. $x \geq y$
- c. $x = y$ or relationship cannot be established
- d. $x < y$
- e. $x \leq y$

Explanations:**1. Questions****School A:**

The number of boys in school A = 120

The number of girls in school A = 180

The total number of students in school A = $120 + 180 = 300$

School B:

The number of boys in school B = 240

The number of girls in school B = 160

The total number of students in school B = $240 + 160 = 400$

School C:

The number of boys in school C = 220

The number of girls in school C = 180

The total number of students in school C = $220 + 180 = 400$

School D:

The number of boys in school D = 170

The number of girls in school D = 130

The total number of students in school D = $170 + 130 = 300$

School E:

The number of boys in school E = 260

The number of girls in school E = 140

The total number of students in school E = $260 + 140 = 400$

School	The number of boys	The number of girls	The total number of students
A	120	180	300
B	240	160	400
C	220	180	400
D	170	130	300
E	260	140	400

Answer: B

The total students in School B = 400.

The total number of participating students inter-school sports events = 45% of 400 = 180.

The number of participating girls inter-school sports events = 40% of 180 = 72. (Since 60% are boys, 40% are girls).

The total girls in School B = 160.

The number of non-participating girls = 160 - 72 = 88

Hence, the correct answer is option B.

2. Questions

School A:

The number of boys in school A = 120

The number of girls in school A = 180

The total number of students in school A = 120 + 180 = 300

School B:

The number of boys in school B = 240

The number of girls in school B = 160

The total number of students in school B = 240 + 160 = 400

School C:

The number of boys in school C = 220

The number of girls in school C = 180

The total number of students in school C = 220 + 180 = 400

School D:

The number of boys in school D = 170

The number of girls in school D = 130

The total number of students in school D = 170 + 130 = 300

School E:

The number of boys in school E = 260

The number of girls in school E = 140

The total number of students in school E = 260 + 140 = 400

School	The number of boys	The number of girls	The total number of students
A	120	180	300
B	240	160	400
C	220	180	400
D	170	130	300
E	260	140	400

Answer: A

The total students in School C = $220 + 180 = 400$.

The total students in School F = $400 + (25\% \text{ of } 400) = 400 + 100 = 500$.

The number of boys in School F = $(3/5) * 500 = 300$.

The number of girls in School D = 130

The required difference = $300 - 130 = 170$

Hence, the correct answer is option A.

3. Questions

School A:

The number of boys in school A = 120

The number of girls in school A = 180

The total number of students in school A = $120 + 180 = 300$

School B:

The number of boys in school B = 240

The number of girls in school B = 160

The total number of students in school B = $240 + 160 = 400$

School C:

The number of boys in school C = 220

The number of girls in school C = 180

The total number of students in school C = $220 + 180 = 400$

School D:

The number of boys in school D = 170

The number of girls in school D = 130

The total number of students in school D = $170 + 130 = 300$

School E:

The number of boys in school E = 260

The number of girls in school E = 140

The total number of students in school E = $260 + 140 = 400$

School	The number of boys	The number of girls	The total number of students
A	120	180	300
B	240	160	400
C	220	180	400
D	170	130	300
E	260	140	400

Answer: C

The number of boys in School B = 240

The number of boys in School A = 120

The required Percentage = $(240 / 120) * 100 = 200\%$

Hence, the correct answer is option C.

4. Questions

School A:

The number of boys in school A = 120

The number of girls in school A = 180

The total number of students in school A = $120 + 180 = 300$

School B:

The number of boys in school B = 240

The number of girls in school B = 160

The total number of students in school B = $240 + 160 = 400$

School C:

The number of boys in school C = 220

The number of girls in school C = 180

The total number of students in school C = $220 + 180 = 400$

School D:

The number of boys in school D = 170

The number of girls in school D = 130

The total number of students in school D = $170 + 130 = 300$

School E:

The number of boys in school E = 260

The number of girls in school E = 140

The total number of students in school E = $260 + 140 = 400$

School	The number of boys	The number of girls	The total number of students
A	120	180	300
B	240	160	400
C	220	180	400
D	170	130	300
E	260	140	400

Answer: E

The total number of girls in schools A, B and E together = $180 + 160 + 140 = 480$

The required average = $480/3 = 160$

Hence, the correct answer is option E.

5. Questions**School A:**

The number of boys in school A = 120

The number of girls in school A = 180

The total number of students in school A = $120 + 180 = 300$

School B:

The number of boys in school B = 240

The number of girls in school B = 160

The total number of students in school B = $240 + 160 = 400$

School C:

The number of boys in school C = 220

The number of girls in school C = 180

The total number of students in school C = $220 + 180 = 400$

School D:

The number of boys in school D = 170

The number of girls in school D = 130

The total number of students in school D = $170 + 130 = 300$

School E:

The number of boys in school E = 260

The number of girls in school E = 140

The total number of students in school E = $260 + 140 = 400$

School	The number of boys	The number of girls	The total number of students
A	120	180	300
B	240	160	400
C	220	180	400
D	170	130	300
E	260	140	400

Answer: D

The number of boys in School B = 240

The number of girls in School E = 140

The required ratio = $240 : 140 = 12 : 7$

Hence, the correct answer is option D.

6. Questions

Company P:

The number of female employees in company P = 128

The number of male employees in company P = $128 + 32 = 160$

The total number of employees in company P = $128 + 160 = 288$

Company Q:

The number of female employees in company Q = 132

The number of male employees in company Q = $132 + 28 = 160$

The total number of employees in company Q = $132 + 160 = 292$

Company R:

The number of female employees in company R = 144

The number of male employees in company R = $144 + 46 = 190$

The total number of employees in company R = $144 + 190 = 334$

Company S:

The number of female employees in company S = 156

The number of male employees in company S = $156 + 24 = 180$

The total number of employees in company S = $156 + 180 = 336$

Company T:

The number of female employees in company T = 115

The number of male employees in company T = $115 + 35 = 150$

The total number of employees in company T = $115 + 150 = 265$

Company	The number of male employees	The number of female employees	The total number of employees
P	160	128	288
Q	160	132	292
R	190	144	334
S	180	156	336
T	150	115	265

Answer: B

The number of male employees in company U = $160 * 125/100 = 200$

The number of female employees in company U = 156

The total number of employees in company U = $200 + 156 = 356$

Hence, the correct answer is option B.

7. Questions

Company P:

The number of female employees in company P = 128

The number of male employees in company P = $128 + 32 = 160$

The total number of employees in company P = $128 + 160 = 288$

Company Q:

The number of female employees in company Q = 132

The number of male employees in company Q = $132 + 28 = 160$

The total number of employees in company Q = $132 + 160 = 292$

Company R:

The number of female employees in company R = 144

The number of male employees in company R = $144 + 46 = 190$

The total number of employees in company R = $144 + 190 = 334$

Company S:

The number of female employees in company S = 156

The number of male employees in company S = $156 + 24 = 180$

The total number of employees in company S = $156 + 180 = 336$

Company T:

The number of female employees in company T = 115

The number of male employees in company T = $115 + 35 = 150$

The total number of employees in company T = $115 + 150 = 265$

Company	The number of male employees	The number of female employees	The total number of employees
P	160	128	288
Q	160	132	292
R	190	144	334
S	180	156	336
T	150	115	265

Answer: A

The difference between the number of female employees in companies R and Q = $144 - 132 = 12$

The difference between the number of male employees in companies S and Q = $180 - 160 = 20$

The required ratio = $12 : 20 = 3 : 5$

Hence, the correct answer is option A.

8. Questions

Company P:

The number of female employees in company P = 128

The number of male employees in company P = $128 + 32 = 160$

The total number of employees in company P = $128 + 160 = 288$

Company Q:

The number of female employees in company Q = 132

The number of male employees in company Q = $132 + 28 = 160$

The total number of employees in company Q = $132 + 160 = 292$

Company R:

The number of female employees in company R = 144

The number of male employees in company R = $144 + 46 = 190$

The total number of employees in company R = $144 + 190 = 334$

Company S:

The number of female employees in company S = 156

The number of male employees in company S = $156 + 24 = 180$

The total number of employees in company S = $156 + 180 = 336$

Company T:

The number of female employees in company T = 115

The number of male employees in company T = $115 + 35 = 150$

The total number of employees in company T = $115 + 150 = 265$

Company	The number of male employees	The number of female employees	The total number of employees
P	160	128	288
Q	160	132	292
R	190	144	334
S	180	156	336
T	150	115	265

Answer: E

The number of male employees in company R = 190

The number of male employees in company T = 150

The required percentage = $(190 - 150) * 100/150 = 26.67\%$ more

Hence, the correct answer is option E.

9. Questions**Company P:**

The number of female employees in company P = 128

The number of male employees in company P = $128 + 32 = 160$

The total number of employees in company P = $128 + 160 = 288$

Company Q:

The number of female employees in company Q = 132

The number of male employees in company Q = $132 + 28 = 160$

The total number of employees in company Q = $132 + 160 = 292$

Company R:

The number of female employees in company R = 144

The number of male employees in company R = $144 + 46 = 190$

The total number of employees in company R = $144 + 190 = 334$

Company S:

The number of female employees in company S = 156

The number of male employees in company S = $156 + 24 = 180$

The total number of employees in company S = $156 + 180 = 336$

Company T:

The number of female employees in company T = 115

The number of male employees in company T = $115 + 35 = 150$

The total number of employees in company T = $115 + 150 = 265$

Company	The number of male employees	The number of female employees	The total number of employees
P	160	128	288
Q	160	132	292
R	190	144	334
S	180	156	336
T	150	115	265

Answer: D

The number of female employees in company P = 128

The number of female employees in company T = 115

The required difference = $128 - 115 = 13$

Hence, the correct answer is option D.

10. Questions

Company P:

The number of female employees in company P = 128

The number of male employees in company P = $128 + 32 = 160$

The total number of employees in company P = $128 + 160 = 288$

Company Q:

The number of female employees in company Q = 132

The number of male employees in company Q = $132 + 28 = 160$

The total number of employees in company Q = $132 + 160 = 292$

Company R:

The number of female employees in company R = 144

The number of male employees in company R = $144 + 46 = 190$

The total number of employees in company R = $144 + 190 = 334$

Company S:

The number of female employees in company S = 156

The number of male employees in company S = $156 + 24 = 180$

The total number of employees in company S = $156 + 180 = 336$

Company T:

The number of female employees in company T = 115

The number of male employees in company T = $115 + 35 = 150$

The total number of employees in company T = $115 + 150 = 265$

Company	The number of male employees	The number of female employees	The total number of employees
P	160	128	288
Q	160	132	292
R	190	144	334
S	180	156	336
T	150	115	265

Answer: C

The total number of employees in companies P and T together = $288 + 265 = 553$

The required average = $553/2 = 276.5$

So,

$$2s = 276.5$$

$$s = 138.25$$

Hence, the correct answer is option C.

11. Questions

The distance between the city P and Q = 350

The speed of the car A = $350/7 = 50$ km/h

The speed of the car B = $50 * 6/5 = 60$ km/h

The speed of the car C = $60 * 125/100 = 75$ km/h

Answer: A

The original time taken by car A to cross the distance = 7 hours

The new speed of the car A = $50 + 20 = 70$ km/h

The new time = $350 / 70 = 5$ hours

The required answer = $7 - 5 = 2$ hours

Hence, the correct answer is option A.

12. Questions

The distance between the city P and Q = 350

The speed of the car A = $350/7 = 50$ km/h

The speed of the car B = $50 * 6/5 = 60$ km/h

The speed of the car C = $60 * 125/100 = 75$ km/h

Answer: C

The time taken by car A to travel 450 km = $450 / 50 = 9$ hours

The time taken by car B to travel 450 km = $450 / 60 = 7.5$ hours

The required ratio = $9 : 7.5 = 6 : 5$

Hence, the correct answer is option C.

13. Questions

The distance between the city P and Q = 350

The speed of the car A = $350/7 = 50$ km/h

The speed of the car B = $50 * 6/5 = 60$ km/h

The speed of the car C = $60 * 125/100 = 75$ km/h

Answer: E

The distance between city P and city Q = 350 km

The speed of car C = 75 km/h

The required time = $350 / 75 = 14/3$ hours

Hence, the correct answer is option E.

14. Questions

Answer: A

The simple interest for 2 years = $5600 - 4800 = 800$

The simple interest for 1 year = $800/2 = 400$

The simple interest for 3 years = $400 * 3 = 1200$

The principal = $4800 - 1200 = 3600$

The rate of simple interest = $400 * 100/3600 = 11.11\%$

Hence, the correct answer is option A.

15. Questions

Answer: B

The initial quantity of milk = $90 * 7/9 = 70$ litres

The initial quantity of water = $90 * 2/9 = 20$ litres

The removed quantity of milk = $27 * 7/9 = 21$ litres

The removed quantity of water = $27 * 2/9 = 6$ litres

The final quantity of milk = $70 - 21 = 49$ litres

The final quantity of water = $20 - 6 + 21 = 35$ litres

The required ratio = $49 : 35 = 7 : 5$

Hence, the correct answer is option B.

16. Questions

Answer: C

Let the marked price of the article = $100a$

The selling price of the article = $100a * 85\% = 85a$

According to the question,

$$85a = 680$$

$$a = 8$$

The marked price of the article = $100 * 8 = \text{Rs. } 800$

The cost price of the article = $800 * 3/4 = \text{Rs. } 600$

Hence, the correct answer is option C.

17. Questions

Answer: D

Let the total capacity of the tank = 90 units [LCM of 15 and 18]

The efficiency of pipe A = $90/15 = 6$ units per minute

The efficiency of pipe B = $90/18 = 5$ units per minute

The total efficiency of A and B together = $6 + 5 = 11$ units per minute

The required time = $90/11 = 8 \frac{2}{11}$ minutes

Hence, the correct answer is option D.

18. Questions

Answer: E

The total investment of P = $18000 * 12$

The total investment of Q = $21000 * 12$

The total investment of R = $24000 * 12$

The profit sharing ratio of P, Q and R = $(18000 * 12) : (21000 * 12) : (24000 * 12) = 6 : 7 : 8$

The profit share of Q = $63000 * \frac{7}{21} = \text{Rs. } 21000$

Hence, the correct answer is option E.

19. Questions

Answer: A

Let the present age of A = $4a$

The present age of B = $5a$

According to the question,

$$(4a + 6) / (5a + 6) = 5 / 6$$

$$24a + 36 = 25a + 30$$

$$a = 6$$

The present age of A = $4 * 6 = 24$ years

The present age of B = $5 * 6 = 30$ years

The average age of A and B = $(24 + 30)/2 = 27$ years

The age of C = $27 + 4 = 31$ years

Hence, the correct answer is option A.

20. Questions

Answer: B

The area of the square = side^2

Let the side of the square = a

According to the question,

$$a^2 = 484$$

$$a = 22$$

The length of the rectangle = The side of the square = 22 cm

The breadth of the rectangle = $22 * 50\% = 11$ cm

The perimeter of the rectangle = $2 (\text{length} + \text{breadth}) = 2 (22 + 11) = 66$ cm

Hence, the correct answer is option B.

21. Questions

Answer: D

$$13\% \text{ of } 400 + ? = 23^2 - 17 * 12 + 19$$

$$52 + ? = 529 - 204 + 19$$

$$52 + ? = 344$$

$$? = 292$$

Hence, the correct answer is option D.

22. Questions

Answer: A

$$5/14 * 980 + 150 = ? + 15 * 24$$

$$350 + 150 = ? + 360$$

$$500 = ? + 360$$

$$? = 140$$

Hence, the correct answer is option A.

23. Questions

Answer: C

$$19^2 + \sqrt{1296} - 150 = ? + 16.66\% \text{ of } 72$$

$$361 + 36 - 150 = ? + (1/6) * 72$$

$$247 = ? + 12$$

$$? = 235$$

Hence, the correct answer is option C.

24. Questions

Answer: E

$$35\% \text{ of } 800 + 16^2 + ? = 5/8 * 1024$$

$$280 + 256 + ? = 640$$

$$536 + ? = 640$$

$$? = 104$$

Hence, the correct answer is option E.

25. Questions

Answer: A

$$7/9 * 729 + 45 * 12 = ? + 145$$

$$567 + 540 = ? + 145$$

$$962 = ?$$

Hence, the correct answer is option A.

26. Questions

Answer: B

$$35.11\% \text{ of } 979.99 + 935.99 / 12.99 = ?$$

$$35\% \text{ of } 980 + 936 / 13 = ?$$

$$343 + 72 = ?$$

$$415 = ?$$

Hence, the correct answer is option B.

27. Questions

Answer: A

$$4.99 * 79.99 + 3.99 * 59.99 = ? + 64.99\% \text{ of } 399.99$$

$$5 * 80 + 4 * 60 = ? + 65\% \text{ of } 400$$

$$400 + 240 = ? + 260$$

$$640 = ? + 260$$

$$380 = ?$$

Hence, the correct answer is option A.

28. Questions

Answer: C

$$44.99\% \text{ of } 199.99 = ? - 19.99 * 9.99 + 24.99\% \text{ of } 799.99$$

$$45\% \text{ of } 200 = ? - 20 * 10 + 25\% \text{ of } 800$$

$$90 = ? - 200 + 200$$

$$90 = ?$$

Hence, the correct answer is option C.

29. Questions

Answer: D

$$39.99 * 14.99 - 24.99 * 11.99 = ? - 13.99 * 15.01$$

$$40 * 15 - 25 * 12 = ? - 14 * 15$$

$$600 - 300 = ? - 210$$

$$300 = ? - 210$$

$$510 = ?$$

Hence, the correct answer is option D.

30. Questions

Answer: E

$$12.21\% \text{ of } 799.99 + 87.29\% \text{ of } 599.98 = ?$$

$$12\% \text{ of } 800 + 87\% \text{ of } 600 = ?$$

$$96 + 522 = ?$$

$$618 = ?$$

Hence, the correct answer is option E.

31. Questions

Answer: A

$$12 * 1 = 12$$

$$12 * 1.5 = 18$$

$$18 * 2 = 36$$

$$36 * 2.5 = 90$$

$$90 * 3 = \mathbf{270}$$

Missing number: 270

Hence, the correct answer is option A.

32. Questions

Answer: B

$$5 * 1 + 1 = 6$$

$$6 * 2 + 2 = 14$$

$$14 * 3 + 3 = 45$$

$$45 * 4 + 4 = \mathbf{184}$$

$$184 * 5 + 5 = 925$$

Missing number: 184

Hence, the correct answer is option B.

33. Questions

Answer: C

$$5 * 2 + 1 = 11$$

$$11 * 2 + 1 = 23$$

$$23 * 2 + 1 = 47$$

$$47 * 2 + 1 = 95$$

$$95 * 2 + 1 = \mathbf{191}$$

Missing number: 191

Hence, the correct answer is option C.

34. Questions

Answer: D

$$32 + 2^2 = 36$$

$$36 + 3^2 = 45$$

$$45 + 4^2 = 61$$

$$61 + 5^2 = 86$$

$$86 + 6^2 = \mathbf{122}$$

Missing number: 122

Hence, the correct answer is option D.

35. Questions

Answer: E

$$18 + 9 = 27$$

$$27 + 18 = 45$$

$$45 + 27 = 72$$

$$72 + 36 = 108$$

$$108 + 45 = \mathbf{153}$$

Missing number: 153

Hence, the correct answer is option E.

36. Questions

Answer: B

$$4 * 3 = 12$$

$$12 * 3 = 36$$

$$36 * 3 = 108$$

$$108 * 3 = 324$$

$$324 * 3 = \mathbf{972}$$

Wrong number: 970

Correct number: 972

Hence, the correct answer is option B.

37. Questions

Answer: A

$$20 * 1.5 = 30$$

$$30 * 2 = 60$$

$$60 * 2.5 = \mathbf{150}$$

$$150 * 3 = 450$$

$$450 * 3.5 = 1575$$

Wrong number: 155

Correct number: 150

Hence, the correct answer is option A.

38. Questions

Answer: E

$$50 + 2 = 52$$

$$52 - 3 = 49$$

$$49 + 5 = 54$$

$$54 - 7 = 47$$

$$47 + 11 = \mathbf{58}$$

Wrong number: 59

Correct number: 58

Hence, the correct answer is option E.

39. Questions**Answer: C**

$$43 + 21 = 64$$

$$64 + 23 = 87$$

$$87 + 25 = 112$$

$$112 + 27 = \mathbf{139}$$

$$139 + 29 = 168$$

Wrong number: 140

Correct number: 139

Hence, the correct answer is option C.

40. Questions**Answer: D**

$$5 * 2 = 10$$

$$10 * 3 = 30$$

$$30 * 4 = 120$$

$$120 * 5 = 600$$

$$600 * 6 = \mathbf{3600}$$

Wrong number: 3500

Correct number: 3600

Hence, the correct answer is option D.

41. Questions**Answer: A**

I). $x^2 - 27x + 182 = 0$

$$x^2 - 14x - 13x + 182 = 0$$

$$x(x - 14) - 13(x - 14) = 0$$

$$(x - 13)(x - 14) = 0$$

$$x = 13 \text{ or } 14$$

II). $y^2 - 21y + 108 = 0$

$$y^2 - 12y - 9y + 108 = 0$$

$$y(y - 12) - 9(y - 12) = 0$$

$$(y - 9)(y - 12) = 0$$

$$y = 9 \text{ or } 12$$

$$\text{So, } x > y$$

Hence, the correct answer is option A.

42. Questions

Answer: D

$$\text{I). } x^2 - 16x + 63 = 0$$

$$x^2 - 9x - 7x + 63 = 0$$

$$x(x - 9) - 7(x - 9) = 0$$

$$(x - 7)(x - 9) = 0$$

$$x = 7 \text{ or } 9$$

$$\text{II). } y^2 - 30y + 209 = 0$$

$$y^2 - 11y - 19y + 209 = 0$$

$$y(y - 11) - 19(y - 11) = 0$$

$$(y - 11)(y - 19) = 0$$

$$y = 11 \text{ or } 19$$

$$\text{So, } x < y$$

Hence, the correct answer is option D.

43. Questions

Answer: B

$$\text{I). } x^2 - 28x + 195 = 0$$

$$x^2 - 15x - 13x + 195 = 0$$

$$x(x - 15) - 13(x - 15) = 0$$

$$(x - 13)(x - 15) = 0$$

$$x = 13 \text{ or } 15$$

$$\text{II). } y^2 - 23y + 130 = 0$$

$$y^2 - 13y - 10y + 130 = 0$$

$$y(y - 13) - 10(y - 13) = 0$$

$$(y - 10)(y - 13) = 0$$

$$y = 10 \text{ or } 13$$

$$\text{So, } x \geq y$$

Hence, the correct answer is option B.

44. Questions

Answer: E

$$\text{I). } x^2 - 31x + 228 = 0$$

$$x^2 - 19x - 12x + 228 = 0$$

$$x(x - 19) - 12(x - 19) = 0$$

$$(x - 12)(x - 19) = 0$$

$$x = 12 \text{ or } 19$$

$$\text{II). } y^2 - 40y + 399 = 0$$

$$y^2 - 21y - 19y + 399 = 0$$

$$y(y - 21) - 19(y - 21) = 0$$

$$(y - 19)(y - 21) = 0$$

$$y = 19 \text{ or } 21$$

$$\text{So, } x \leq y$$

Hence, the correct answer is option E.

45. Questions

Answer: C

$$\text{I). } x^3 = 216$$

$$x = 6$$

$$\text{II). } y^2 - 12y + 36 = 0$$

$$y^2 - 6y - 6y + 36 = 0$$

$$y(y - 6) - 6(y - 6) = 0$$

$$(y - 6)(y - 6) = 0$$

$$y = 6 \text{ or } 6$$

So, $x = y$ or relationship cannot be established.

Hence, the correct answer is option C.